MCO — METACHRONAL CASCADE ORNITHOPTER

Master Design Specification · Issue 1.1 · 2026-06-03

Updates from Issue 1.0: clamshell fairings moved to provisional; rib specification reviewed and updated to Rev H (wall 1.0mm); MCO-1A rib construction locked.

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1. Programme Overview

The MCO programme develops a forward-flight ornithopter using an 8-wing metachronal cascade. Hover is not a design requirement on either vehicle. The MCO-1D is the mandatory flight test programme for the MCO-1A — every wave library mode is validated on the drone before the crewed aircraft is built.

Vehicle	Role	MTOW	Chord	Airfoil	Power
MCO-1D	Drone prototype — validates wave library	3,448 g	165mm	S1223 + VGs + Gurney	6S 5000mAh LiPo
MCO-1A	Piloted SAR — crewed forward flight	376 kg	924mm	FX 63-137 + Gurney	CBR600RR series hybrid

2. MCO-1D Wing Specification (Rev H — updated this issue)

2.1 Wing geometry (unchanged from Rev G)

	W1	W2	W3	W4	Units
Semi-span	772	749	735	712	mm
Tip-to-tip	1,544	1,498	1,470	1,424	mm
Chord	165	165	165	165	mm
AR	4.7	4.5	4.5	4.3	—

2.2 Rib construction (Rev H — wall thickness update)

UPDATE Rev H: Standard rib wall reduced from 1.2mm to 1.0mm. Rationale: ribs are shape-limited not load-limited. Peak load per rib bay in forward flight sprint is 0.79N — trivially within 1.0mm LW-PLA capability. 1.0mm gives 2.5 perimeters with 0.4mm nozzle — reliable print. Mass saving: ~84g across all 280 standard ribs.

Component	Specification	Rationale
Airfoil	S1223 (UIUC exact coordinates)	Optimised for Re 2×10■
D-box zone	LE to 65%c — LW-PLA gyroid 12%	Shape accuracy + low mass
CF skin	100gsm 2×2 twill, 1 ply, upper + lower, LE–65%c	99% of torsional stiffness; aerodynamic surface
Spars (×3)	2.0mm OD / 1.0mm ID pultruded CF at 10%c, 20%c, 55%c	3-spar triangle holds D-box geometry — not just load
D-box rear wall	Full-depth rib element at 65%c, 0.5mm wall	Closes torsion box; anchors tail zone
Rib tail zone	65–100%c, 0.5mm wall LW-PLA, exact S1223 profile	Profile accuracy in separation-critical TE region
Standard rib wall	1.0mm ← Rev H update (was 1.2mm)	Load-limited: 0.79N max; 1.0mm is adequate + saves 84g
Root rib	3.0mm solid LW-PLA (unchanged)	Root installation loads
Tip rib	2.0mm solid LW-PLA (unchanged)	Tip structural integrity
Rib spacing	20mm (17%c)	Membrane deflection 0.03mm at cruise — negligible

2.3 Membrane and tail zone (confirmed, forward-flight rationale)

The membrane is MORE important in forward flight than in hover, not less. Three reasons: (1) profile accuracy — the S1223's reflexed TE manages separation at high CL; any distortion costs CL,max regardless of flight mode. (2) anti-warp — the wing still flaps; the membrane provides spanwise continuity in the tail zone against differential flap loading. (3) surface smoothness — forward flight requires sustained laminar flow in the aft section for good L/D; open rib-tail gaps would trip the BL catastrophically. Removing hover does not justify removing the membrane.

Component	Specification	Status
Membrane	Ripstop nylon 40 denier, E6000 to rib tail edges, 65–100%c	✓ CONFIRMED

Component	Specification	Status
TE rod	0.3mm CF rod, E6000 hem bond — anchors membrane at TE	✓ CONFIRMED
Gurney flap	1.2mm (1%c), integral LW-PLA, lower surface TE	✓ CONFIRMED
Covert flaps	0.125mm PET, 50–62%c upper surface, 3 strips, passive	✓ CONFIRMED — more useful in fwd flight than ho
VGs	17%c, h=8.4mm, 18° toe, counter-rotating pairs, LW-PLA strip	✓ CONFIRMED — essential at Re 2×10■
Raked wingtip	25° LE sweep, 15%c (18mm) extension, 1.5° washout, 5° dihedral	✓ CONFIRMED

3. MCO-1A Wing Specification

3.1 Wing geometry

	W1	W2	W3	W4	Units
Semi-span	4,324	4,199	4,116	3,991	mm
Tip-to-tip	8,648	8,398	8,232	7,982	mm
Chord	924	924	924	924	mm
AR	4.7	4.5	4.5	4.3	—
Airfoil	FX 63-137	FX 63-137	FX 63-137	FX 63-137	Wortmann

Total area: 30.7 m² · Wing loading: 2.0 lb/ft² · CL,max (FX+Gurney): ~2.07 · Stall: 8.7 m/s (17 kt) · Best L/D: ~16:1

3.2 Rib and wing construction (CFRP — locked)

Component	Specification	Notes
Airfoil	FX 63-137 (Wortmann, UIUC coordinates)	Post-NACA, optimised Re 10■–10■
D-box zone	CNC-cut 2mm CFRP sheet ribs, LE to 65%c	±0.2mm accuracy at 924mm chord — adequate
D-box skin	CFRP prepreg 2–3 ply, upper + lower, LE–65%c	Structural torsion box; aerodynamic surface
Spars (×3)	CFRP tube or I-beam at 10%c, 20%c, 55%c + rear wall 65%c	Scale equivalent of drone 3-spar layout
Rib tail zone	3mm CFRP sheet, exact FX 63-137 profile, 65–100%c	Load-bearing at full scale (50N per bay at CL,max)
Tail skin	0.1mm fibreglass prepreg, both surfaces, 65–100%c	REPLACES membrane — membrane inadequate at 50N/bay loading
Rib spacing	157mm (17%c at 924mm chord)	Consistent with drone 17%c
Gurney flap	9.24mm (1%c), CFRP angle section, bonded lower TE	Valid at any Re — purely mechanical TE effect
Covert flaps	0.125mm PET or thin CFRP, 12%c length (~111mm), passive	Deploy at high AoA — more useful in fwd flight
VGs	NONE	Counterproductive at Re >5×10■ — removed
Raked wingtip	25°/15%c (139mm extension) — Re 10■ geometry review req.	Valid as span extension; exact geometry TBD for Re 10■

3.3 Wing stow — provisional note

PROVISIONAL NOTE: Spring-loaded CFRP clamshell root fairings (one per wing station × 8) — REMOVED from locked specification pending further examination. The fairings were specified to close over stowed wings at 210°, reducing stowed wing drag area from ~1.6 m² to ~0.02 m². They remain the correct engineering solution for high-speed sprint modes and are likely to be incorporated in the final build. However the fabrication complexity, mass (~3.4 kg), and spring-force sizing relative to sweep servo torque need detailed design work before committing. Open item O-F1: fairing design, spring sizing, and servo torque verification. Without fairings: sprint modes above 68 kt require updated drag assessment.

Stow parameter	Value	Notes
Max stow angle	210°	Hard stop repositioned — Tweak 2 (Rev B)
Hard stop load rating	115N (3× safety factor)	Resists aero blow-open toward 180°
Stow order	W4 → W3 → W2 → W1	Locked in FCC firmware — collision prevention
Stow rate	60°/s minimum	Verified at current span; extended W1+W2 needs check
Tip dihedral at stow	–5° (tips tuck inward)	Reduces tip overhang past fuselage
Root fairings	PROVISIONAL — see note above	O-F1 open item
Active servo galloping	7.0 Hz effective fn with servo active	Replaces retired clip

4. MCO-1A Powertrain — CBR600RR Series Hybrid (locked)

Component	Specification	Mass	Notes
IC engine	Honda CBR600RR stripped (or Kawasaki ZX-6R 636)	~53 kg	119 hp / 89 kW peak; ~70 kW at 60% continuous
Generator	3-phase PMG, direct-coupled to engine output shaft	~8 kg	No PSRU needed — series hybrid, generator at engine RPM
Exhaust + mount	Stainless system, CFRP engine mount	~4 kg	Routed belly aft
Sprint battery	Li-ion 5 kWh (200 Wh/kg cells)	~12 kg	6 min full-sprint reserve; recharges in ~5 min at loiter
Fuel	MOGAS 91, 15L	~11 kg	~45–60 min endurance at cruise
Cooling	Compact radiator + NACA inlet (liquid-cooled CBR)	~3 kg	Modest load at 60% power; BMW boxer alt. = no radiator
Oil system	Standard wet sump + accusump	~0.5 kg	Negative-g prohibited — acceptable for SAR mission
Wing motors ×8	Large brushless, crank-and-rod mechanism	—	Same electrical architecture as MCO-1D, scaled
FCC	RPi Zero 2W + Pixhawk Mini + pitot	—	Identical wave library software — direct transfer from MCO-1D

System mass ~88 kg · MTOW 376 kg · Sprint 136 kt (generator) · 159 kt (+ battery) · Cost ~$5,000–6,000 total

5. MCO-1A Performance Envelope

Mode	Speed	Notes
Stall — all wings deployed	8.7 m/s (17 kt)	MTOW 376 kg, CL,max 2.07
Best endurance	~12 m/s (23 kt)	SAR loiter / thermal soaring
Best range / glide	~16 m/s (31 kt)	L/D ~16:1 (FX 63-137 at Re 10■)
W3+W4 partial stow cruise	~24 m/s (46 kt)	W1+W2 faired at 210°
GA sprint — generator only (135°)	~70 m/s (136 kt)	Cessna 172 class ✓ 70 kW sustained
GA sprint — + battery burst	~82 m/s (159 kt)	Piper Arrow class ✓ ~100 kW for 6 min
STOOP-2A ballistic (aerodynamic terminal)	~80 m/s (156 kt)*	All wings faired at 210° — drag calc only
Best glide (no flap)	~19 m/s (37 kt)	L/D ~16:1
Altitude cruise at 4,000m	30 m/s IAS → 37 m/s TAS	22% more ground/litre than sea level

* Without clamshell fairings (provisional): stoop terminal significantly lower due to stowed wing drag. Fairing design (O-F1) governs this figure.

6. Open Items

#	Item	Priority	Status
O-1	MCO-1D hard stop: reposition collar 200°→210°	Critical	Before sprint/stoop test
O-2	wave_library.py: SWEEP_MAX_DEGREES=210, STOW_ANGLE=210	Critical	2-line change
O-3	MCO-1D flight test T1–T9 (full validation programme)	Critical	After O-1, O-2
O-4	MCO-1D rib reprint at 1.0mm wall (Rev H update)	High	Before first flight
O-F1	MCO-1A: Clamshell fairing design, spring sizing, servo torque check	High	Provisional — design before MCO-1A wing build
O-5	MCO-1A: Galloping re-analysis for extended W1+W2 (if pursued)	High	Before span extension commit
O-6	MCO-1A: Stow servo torque for extended W1+W2 (+69% moment)	High	Before span extension commit
O-7	MCO-1A: Root CFRP layup uprate for extended W1+W2 (+69% bending)	High	Structural design phase
O-8	MCO-1A: Raked tip geometry review at Re 10■	Medium	Before wing design freeze
O-9	MCO-1A: CBR600RR engine procurement and generator matching	Medium	After MCO-1D validation
O-10	MCO-1A: FAA Experimental Airworthiness application	Medium	After MCO-1D complete

7. Document Register

Document	Issue	Contents
MCO_Master_Specification_v2.pdf	1.1 (this)	Consolidated spec — current authority

Document	Issue	Contents
MCO_wave_library_addendum.pdf	Rev A	Tweaks 3+4: variable sweep, dihedral bias
MCO_wave_library_addendum_revB.pdf	Rev B	Tweak 2: 210° stow, fairings (provisional)
MCO_1D_Wing_Construction.pdf	Rev G	Wing build — rib wall update to Rev H pending
MCO_1D_Rib_Drawing.pdf	Rev G → H	Rib geometry — wall 1.2mm→1.0mm update pending
MCO_1D_VG_Design.pdf	Rev E	VG placement and installation

MCO Master Design Specification Issue 1.1 · 2026-06-03 · Changes from Issue 1.0: clamshell fairings provisional (O-F1); rib wall 1.2→1.0mm (Rev H); MCO-1A CFRP rib and fibreglass tail skin locked; membrane confirmed for forward-flight reasons. MCO-1D specification frozen. MCO-1A powertrain (CBR600RR series hybrid) locked.